Week 20 Self-Assessment

Discussion Sections and Interpretation

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1 Instructions

Test your understanding of how to write Discussion sections and interpret research findings. The Discussion is where you make sense of your results and connect them to the broader literature.

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2 Part 1: Conceptual Questions

2.1 Question 1: Purpose of the Discussion

The main purpose of a Discussion section is to:

- (A) Report all statistical values
- (B) Describe the methodology
- (C) Interpret results and explain their meaning in the context of theory and previous research
- (D) List all limitations

2.2 Question 2: Discussion Structure

A Discussion section typically follows this order:

- (A) Limitations, then results summary, then implications
- (B) Results summary, interpretation/connection to literature, limitations, implications/future directions
- (C) Future research, then limitations, then summary
- (D) Any order is acceptable

2.3 Question 3: Results vs Discussion

Statistical values (like *t*, *p*, *r*) belong primarily in:

- (A) The Results section
- (B) The Discussion section
- (C) Both equally
- (D) The Method section

Interpretation and theoretical implications belong primarily in:

- (A) The Results section
- (B) The Discussion section
- (C) The Introduction
- (D) The Abstract

2.4 Question 4: Opening the Discussion

A good opening sentence for the Discussion should:

- (A) Start with limitations
- (B) Summarise the main finding and whether it supported the hypothesis
- (C) Repeat the method
- (D) Introduce new analyses

2.5 Question 5: Connecting to Literature

When discussing findings, you should:

- (A) Only cite studies that agree with your results
- (B) Relate findings to previous research, explaining both consistencies and inconsistencies
- (C) Avoid citing any previous research
- (D) Only mention the studies from your Introduction

2.6 Question 6: Discussing Non-Significant Results

If results are NOT significant, you should:

- (A) Claim they are significant anyway
- (B) Hide the study
- (C) Discuss possible reasons such as low power, measurement issues, or a genuine null effect
- (D) Only write a brief Discussion

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3 Part 2: Interpretation Questions

3.1 Question 7: Interpreting Significant Results

A study finds that caffeine improves reaction time ($p < .05$, $d = 0.65$). In the Discussion, you should:

- (A) Just state 'the hypothesis was supported'
- (B) Explain what this means, compare to previous findings, and discuss practical implications
- (C) Report the statistics again
- (D) Move directly to limitations

3.2 Question 8: Interpreting Effect Sizes

Your study found a significant effect with $d = 0.25$ (small). In the Discussion, you should:

- (A) Claim the effect is large because it's significant
- (B) Acknowledge the effect is statistically significant but small, and discuss what this means practically
- (C) Ignore the effect size
- (D) Say the results are not meaningful

3.3 Question 9: Interpreting Correlational Findings

A correlational study finds r = .45 between X and Y. In the Discussion, you should:

- (A) Conclude that X causes Y
- (B) Conclude that Y causes X
- (C) Discuss the association without claiming causation, and mention possible third variables
- (D) Say the correlation proves the theory

3.4 Question 10: Unexpected Results

Your results contradict your hypothesis. In the Discussion, you should:

- (A) Claim the results actually support your hypothesis
- (B) Hide the contradictory findings
- (C) Acknowledge the unexpected result and explore possible explanations
- (D) Blame participants or methodology without analysis

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4 Part 3: Application - Limitations

4.1 Question 11: What Makes a Good Limitation?

A good limitation is:

- (A) A vague complaint about the study
- (B) A specific issue that could have affected results, with explanation of potential impact
- (C) An excuse for non-significant findings
- (D) A comment about how hard the research was

4.2 Question 12: Types of Limitations

Match each limitation to its category:

"Only university students were sampled" -

- (A) Sample/generalisability
- (B) Measurement
- (C) Design
- (D) Statistical

"Self-report measures may not reflect actual behaviour" -

- (A) Sample

- (B) Measurement
- (C) Design
- (D) Analysis

"The between-subjects design may have been affected by individual differences" -

- (A) Sample
- (B) Measurement
- (C) Design
- (D) Statistical

"The small sample size may have limited power to detect effects" -

- (A) Sample
- (B) Measurement
- (C) Design
- (D) Statistical power

4.3 Question 13: Constructive Limitations

After stating a limitation, you should:

- (A) Move to the next limitation
- (B) Explain how it might have affected results and suggest how future research could address it
- (C) Apologise for the flaw
- (D) Argue why it doesn't matter

4.4 Question 14: Limitations to Avoid

Which is NOT a good limitation to mention?

- (A) Small sample size affecting power
- (B) I found the statistics confusing
- (C) Convenience sampling limiting generalisability
- (D) Self-report measures being subject to bias

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5 Part 4: Application - Future Directions

5.1 Question 15: Suggesting Future Research

Good future directions should:

- (A) Be completely unrelated to your study
- (B) Build logically on your findings and address limitations
- (C) Promise to solve all problems
- (D) Repeat your study exactly

5.2 Question 16: Examples of Future Directions

Which is the best future direction suggestion?

```
opts_future <- c(
  "Future research should do a better study",
  answer = "Future research could examine whether this effect generalises to
clinical populations, as our sample was limited to healthy adults",
  "More research is needed",
  "Future studies should use more participants"
)
```

- (A) Future research should do a better study
- (B) Future research could examine whether this effect generalises to clinical populations, as our sample was limited to healthy adults
- (C) More research is needed
- (D) Future studies should use more participants

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6 Part 5: MCQ Exam Practice

6.1 Scenario 1: Writing a Discussion Opening

A study tested whether exercise improves mood. Results: participants reported significantly better mood after exercise ($p = .003$, $d = 0.72$).

1. **The best opening sentence would be:**

```
opts_open <- c(
  "We did a study on exercise and mood",
  answer = "The present study examined whether exercise improves mood. Results
supported the hypothesis, with participants reporting significantly improved
mood following exercise",
  "The statistics were significant",
  "There were several limitations"
)
```

- (A) We did a study on exercise and mood
- (B) The present study examined whether exercise improves mood. Results supported the hypothesis, with participants reporting significantly improved mood following exercise
- (C) The statistics were significant
- (D) There were several limitations

2. **What should come next?**

- (A) Limitations
- (B) Interpretation and connection to previous research
- (C) More statistics

- (D) Conclusion

3. **When connecting to literature, you might write:**

- (A) These findings are consistent with previous research (Smith, 2020) showing...
- (B) We got good results
- (C) The p-value was significant
- (D) The Method was appropriate

6.2 Scenario 2: Discussing Correlational Findings

A study finds a moderate positive correlation ($r = .48$) between smartphone use and anxiety.

1. **Can we conclude smartphones CAUSE anxiety?** TRUE / FALSE

2. **In the Discussion, we should:**

- (A) Claim causation because the correlation is moderate
- (B) Discuss the association and acknowledge we cannot determine causation
- (C) Ignore the correlation strength
- (D) Only discuss if it was significant

3. **What third variable should we mention?**

- (A) Possible confounds like stress, personality, or life circumstances
- (B) The sample size
- (C) The correlation coefficient
- (D) The degrees of freedom

4. **A limitation related to this design would be:**

- (A) The correlation was too strong
- (B) Correlational design prevents causal conclusions
- (C) Smartphones are hard to measure
- (D) Anxiety is not important

6.3 Scenario 3: Handling Non-Significant Results

A study predicted that sleep deprivation would impair memory. Results: $t(38) = 1.45$, $p = .154$, $d = 0.46$.

1. **Is this significant at alpha = .05?** TRUE / FALSE

2. **Does p = .154 PROVE the null hypothesis is true?** TRUE / FALSE

3. **In the Discussion, appropriate responses include:**

The sample may have been too small (low power): TRUE / FALSE

The measurement may not have been sensitive enough: TRUE / FALSE

The hypothesis was definitely wrong: TRUE / FALSE

The effect size ($d = 0.46$) suggests a medium effect worth investigating: TRUE / FALSE

4. **A constructive approach would be:**

- (A) Claim the results are actually significant
- (B) Abandon this research area
- (C) Acknowledge the non-significant finding, note the medium effect size, and suggest a larger sample in future research
- (D) Blame participants

6.4 Scenario 4: Complete Discussion Elements

Order these elements as they would typically appear in a Discussion:

First:

- (A) Summary of findings and whether hypothesis was supported
- (B) Detailed limitations
- (C) Future research
- (D) Reference list

Second:

- (A) Limitations
- (B) Interpretation and relation to previous research
- (C) Conclusion
- (D) Abstract

Third:

- (A) More results
- (B) Introduction material
- (C) Limitations of the study
- (D) Method details

Fourth:

- (A) Statistics
- (B) Procedure
- (C) Implications and future directions
- (D) Hypothesis

Final:

- (A) More limitations
- (B) Concluding statement
- (C) New analyses
- (D) Additional hypotheses

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7 Part 6: Writing Practice

7.1 Question 17: Discussion Sentence Types

Identify which type of sentence each is:

"The present study found that participants who received feedback performed significantly better than controls." -

- (A) Summary of findings
- (B) Interpretation
- (C) Limitation
- (D) Future direction

"This finding is consistent with Self-Determination Theory (Deci & Ryan, 2000), which suggests…" -

- (A) Summary
- (B) Interpretation/theory connection
- (C) Limitation
- (D) Future direction

"A limitation of this study is the reliance on self-report measures, which may be subject to social desirability bias." -

- (A) Summary
- (B) Interpretation
- (C) Limitation
- (D) Future direction

"Future research could examine whether these effects persist over longer time periods." -

- (A) Summary
- (B) Interpretation
- (C) Limitation
- (D) Future direction

7.2 Question 18: Avoiding Common Errors

Which sentence is PROBLEMATIC for a Discussion?

- (A) The hypothesis was supported
- (B) The results prove that X causes Y (when using correlational design)
- (C) The effect size was medium
- (D) Results were consistent with previous findings

7.3 Question 19: Practical vs Statistical Significance

A study finds $p = .01$, $d = 0.15$. The Discussion should:

- (A) Focus only on the significant p-value
- (B) Ignore the effect size

- (C) Note that while statistically significant, the effect is small and may have limited practical importance
- (D) Claim the effect is large because it's significant

8 Summary Checklist

Before writing your Discussion section, verify you can:

- ☐ Summarise main findings and whether hypothesis was supported
- ☐ Interpret results in context of theory and previous research
- ☐ Discuss both consistencies and inconsistencies with literature
- ☐ Identify specific, meaningful limitations
- ☐ Explain how limitations might have affected results
- ☐ Suggest concrete future research directions
- ☐ Avoid claiming causation from correlational data
- ☐ Appropriately discuss non-significant findings
- ☐ Balance statistical and practical significance

> ! Key Points for Your Lab Report
>
> Your Discussion should tell a story: What did you find? What does it mean? What are the limitations? Where do we go from here? Avoid simply repeating statistics - instead, explain and interpret them.